# Lazy ring + one directed shortcut: Bimodality at $K = 2$ vs. $K = 4$ and the jump-over mechanism (Exact FPT via AW inversion + trajectory decomposition)

## 1 Problem restatement and objective

In the lazy ring with one directed shortcut (selfloop rule) we previously observed (for $K = 2$, i.e. only $\pm 1$ moves) that a double peak in the first-passage/first-absorption pmf $f(t)$ only appears for sufficiently small shortcut strength $\beta$. As $\beta$ increases, the shortcut is taken more often, mass concentrates on a faster time scale, and the slow peak is suppressed, so bimodality disappears.

Here we fix the start node to $n_0$ and set the absorbing target to the "farthest" (approximately antipodal) node, then scan the $K = 4$ case (allowed ring moves $\pm 1, \pm 2$). The key new feature at $K = 4$ is that a single ring move can *jump over* the absorbing target without landing on it (e.g. from target $- 1$ to target $+ 1$), which may alter the bimodal mechanism.

The goal is two-layered:

- **Layer A (where bimodality occurs):** use an exact/high-precision method to scan parameters and apply one fixed bimodality criterion to flag true double peaks.
- **Layer B (why bimodality occurs):** for representative bimodal cases, decompose trajectories into *mutually exclusive* classes based on shortcut usage and jump-over events, and quantify how those classes populate the first peak / valley / second peak.

## 2 Model definition (fixed convention)

### 2.1 State space and ring moves

We use paper-style indexing $1, \ldots, N$ with all additions taken modulo $N$. Let $K = 2k$ be the ring degree. At each step, with probability $q$ the walker makes a ring move and with probability $1 - q$ it stays (lazy self-loop). Given that a ring move occurs, the displacement is chosen uniformly from $\{\pm 1, \ldots, \pm k\}$; hence each ring edge has probability $q/K$.

### 2.2 One directed shortcut (selfloop rule)

We add a single directed shortcut src $\to$ dst. Under the *selfloop rule*, at the source we take probability mass from the self-loop and reassign it to the shortcut:

$$p = \beta(1 - q), \qquad 0 \le p \le 1 - q \quad (\Leftrightarrow\ 0 \le \beta \le 1).$$

Explicit one-step probabilities:

- for $i \neq \text{src}$:
$$P(i \to i) = 1 - q, \qquad P(i \to i \pm r) = \frac{q}{K}, \quad r = 1, \ldots, k;$$
- for $i = \text{src}$:
$$P(\text{src} \to \text{src}) = 1 - q - p, \quad P(\text{src} \to \text{dst}) = p, \quad P(\text{src} \to \text{src} \pm r) = \frac{q}{K}.$$

### 2.3 Absorption and first-absorption time

When a step ends at target, absorption occurs with probability $\rho$; otherwise the walker survives and continues from target. In this report we take $\rho = 1$ (strict first passage), so the first-absorption time $T$ equals the first hitting time of target.

### 2.4 Geometry (paper-like, with antipodal target)

We fix

$$n_0 = 1, \qquad \text{target} = n_0 + \left\lfloor \frac{N}{2} \right\rfloor, \qquad \text{src} = n_0 + 5, \qquad \text{dst} = \text{target} + 1,$$

with modular arithmetic on the ring. The choice dst = target + 1 avoids the special "shortcut points directly to the absorber" mechanism (dst = target), which is discussed separately in `valley_dst_report`.

### 2.5 Relation to other reports in this repo

Other reports (e.g. `valley_report` / `valley_dst_report`) also use a "add an outgoing edge at src and renormalize" construction, often in a non-lazy setting. That convention differs from the present selfloop rule (which preserves the ring-move probability $q$). Accordingly, in this report all scans and statistics refer to the **lazy + selfloop** convention unless explicitly stated.

## 3 Bimodality criteria (fixed, paper-style)

We use the same criterion as in existing reports:

- **Strict local peak:** $f(t) > f(t-1)$, $f(t) > f(t+1)$, and $f(t) \geq h_{\min}$, with boundary convention $f(0) = f(T+1) = 0$.
- **Paper bimodality:** at least two peaks, and the 2nd-highest peak height $\geq 1\%$ of the highest.
- **Macro bimodality (optional):** additionally require $t_2/t_1 \geq 10$ for the two dominant peaks.

Defaults: $h_{\min} = 10^{-7}$, relative-height threshold 1%, macro ratio threshold 10.

## 4 Two path events and mutually-exclusive classes

### 4.1 Shortcut traversal count $C$

Let $C$ be the number of times a trajectory traverses the directed edge src $\to$ dst. We use the coarse indicator $C \geq 1$ (and also record the $0/1/\geq 2$ breakdown for robustness).

### 4.2 Jump-over count $J$ (new at $K = 4$)

For $K = 4$ (allowed $\pm 2$ ring moves), a ring step can cross the target without landing. We count jump-over events **only on ring moves** (not on self-loops or shortcut steps). A step from $x$ to $x \pm r$ ($r \geq 2$) is a jump-over if the target lies strictly between departure and arrival along that direction:

$$\text{target} \in \{x \pm 1, \ldots, x \pm (r-1)\}.$$

Equivalently, for a $+r$ move define the directed ring distance $d_+(x, \text{target}) = (\text{target} - x) \bmod N$; jump-over occurs iff $1 \leq d_+ \leq r-1$ (and similarly for $-r$ using $d_-(x, \text{target}) = (x - \text{target}) \bmod N$). For $K = 2$ we always have $J \equiv 0$.

### 4.3 Four mutually-exclusive classes (A/B/C/D)

Using the indicators $(C \geq 1)$ and $(J \geq 1)$ we define:

$$\text{A: } (C = 0, J = 0), \quad \text{B: } (C \geq 1, J = 0), \quad \text{C: } (C = 0, J \geq 1), \quad \text{D: } (C \geq 1, J \geq 1),$$

which are mutually exclusive and exhaustive. For $K = 2$, the classification degenerates to shortcut vs. nonshortcut.

## 5 Peak–valley–peak windows and conditional statistics

Given an exact $f(t)$, we locate the first two peaks in time order $(t_1 < t_2)$ under the paper criterion, and define the valley bottom

$$t_v = \arg\min_{t_1 < t < t_2} f(t).$$

We set a symmetric window half-width

$$\Delta = \max\left(1, \lfloor 0.05\,(t_2 - t_1) \rfloor\right),$$

and label each Monte Carlo trajectory by the nearest center among $\{t_1, t_v, t_2\}$ if it lies within distance $\Delta$ (ties are assigned to "other"). For each window $W$ we report:

- window mass $\Pr(T \in W)$;
- $\Pr(C \geq 1 \mid T \in W)$ and $\Pr(J \geq 1 \mid T \in W)$ (Wilson 95% intervals recorded in `*.summary.json`);
- class fractions for A/B/C/D.

## 6 Exact $f(t)$ computation: AW inversion vs. time-domain iteration

### 6.1 AW inversion for the main model

For the main model (point absorption at target, no "jumpover absorbs" modification), we compute the generating function and invert it via Abate–Whitt (AW).

**Defect-free lazy ring (spectral form)** The defect-free eigenvalues are

$$\lambda_\ell = (1 - q) + \frac{2q}{K} \sum_{r=1}^{k} \cos\left(\frac{2\pi r \ell}{N}\right), \qquad \ell = 0, \ldots, N-1.$$

Define the translation-invariant resolvent

$$Q(d, z) = \frac{1}{N} \sum_{\ell=0}^{N-1} \frac{\cos(2\pi \ell d/N)}{1 - z\lambda_\ell}, \qquad |z| < 1,$$

where $d$ is the ring displacement.

**Rank-one defect (selfloop shortcut) and point absorption** The selfloop shortcut modifies only the outgoing probabilities at src, i.e. a single-column rank-one defect relative to the defect-free ring. We evaluate the corresponding resolvent entries $S_{i,j}(z)$ using a Sherman–Morrison update expressed in terms of $Q(\cdot, z)$. For imperfect absorption $\rho \in (0, 1]$, the first-absorption generating function is

$$\tilde{f}(z) = \sum_{t \geq 1} f(t) z^t = \frac{\rho\, S_{\text{target},n_0}(z)}{(1 - \rho) + \rho\, S_{\text{target,target}}(z)}.$$

For $\rho = 1$, $\tilde{f}(z) = S_{\text{target},n_0}(z) / S_{\text{target,target}}(z)$.

**AW (FFT) coefficient recovery** We sample $z_m = re^{2\pi i m/L}$ and approximate

$$f(t) \approx r^{-t} \frac{1}{L} \sum_{m=0}^{L-1} \tilde{f}(z_m)\, e^{-2\pi i m t/L}, \qquad t = 1, \ldots, T_{\max},$$

implemented via FFT. We use $r = 10^{-4/T_{\max}}$ and $L$ the next power of two exceeding $T_{\max}$ by a small pad, consistent with other AW scripts in this repo. For the representative plots we take $T_{\max} = 4000$ to cover both dominant peaks and the valley.

### 6.2 Time-domain master-equation iteration (scans and nonlocal control)

Layer A scans require suppressing tail mass (e.g. survival $\lesssim 10^{-14}$), so we use time-domain master-equation iteration (absorbing flux recursion) to obtain numerically exact $f(t)$ for the fixed bimodality criterion. The counterfactual control "jumpover $\to$ absorb" makes absorption a nonlocal event (crossing without landing), so it is also computed in the time domain rather than by the point-absorption AW formula.

## 7 Layer A: where bimodality truly occurs (parameter scans)

### 7.1 $N$ scan at fixed $\beta = 0.02$

Fix $q = 2/3$, $\beta = 0.02$ (so $p = \beta(1-q)$), and the geometry above. Scan even $N = 10..200$:

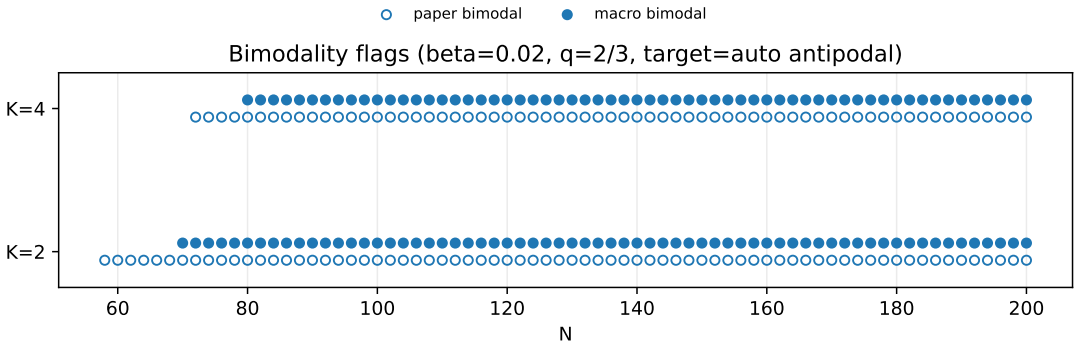


Figure 1: Bimodality flags for $\beta = 0.02$ at $K = 2$ vs. $K = 4$ (hollow = paper, filled = macro).

### 7.2 $\beta$ scan at fixed $N = 100$

Fix $N = 100$, $q = 2/3$ and scan $\beta$:

## 8 Layer B: mechanism via trajectory decomposition

### 8.1 Representative point: $N = 100, \beta = 0.02$ ($K = 2$ vs. $K = 4$)

We first show the exact $f(t)$ (log scale) with peak/valley/peak markers:

Using the same window definition, we show window-conditional class fractions:

#### 8.1.1 Quantitative window summaries

To make the mechanism claims auditable, we tabulate the window-conditional statistics (generated from `*.summary.json`):

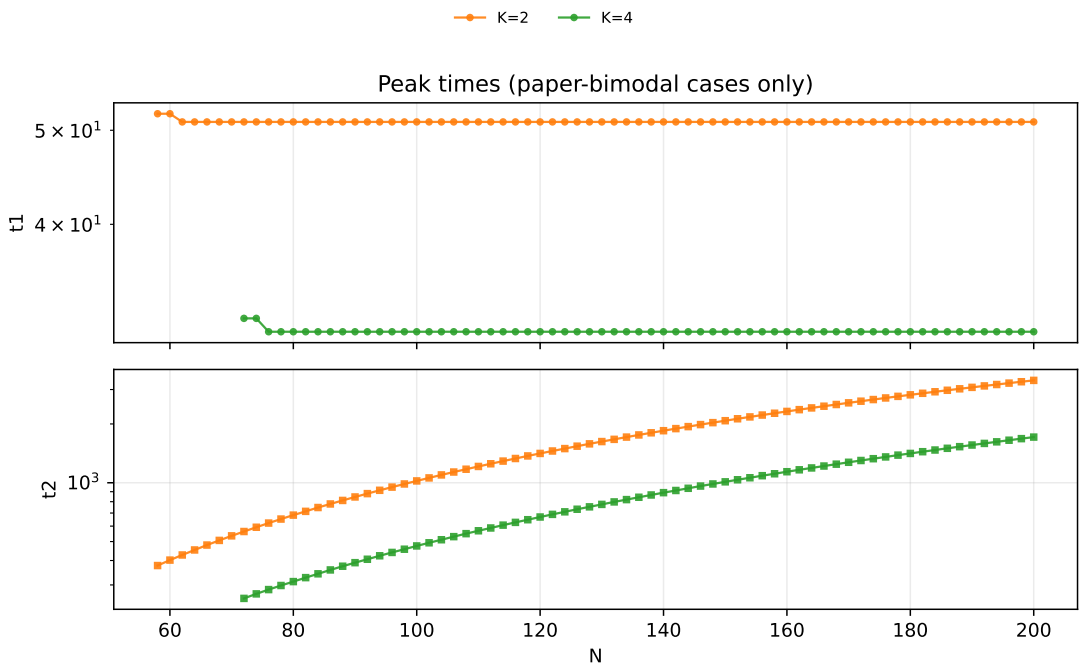


Figure 2: Peak times $t_1, t_2$ (log scale), restricted to paper-bimodal cases.

| $\beta$ | $p$ | paper | macro | $t_1$ | $t_2$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | – | – |
| 0.0005 | 0.000166667 | 1 | 1 | 54 | 1243 |
| 0.001 | 0.000333333 | 1 | 1 | 54 | 1236 |
| 0.002 | 0.000666667 | 1 | 1 | 54 | 1222 |
| 0.005 | 0.00166667 | 1 | 1 | 53 | 1183 |
| 0.01 | 0.00333333 | 1 | 1 | 53 | 1124 |
| 0.02 | 0.00666667 | 1 | 1 | 51 | 1022 |
| 0.03 | 0.01 | 1 | 1 | 50 | 935 |
| 0.05 | 0.0166667 | 1 | 1 | 48 | 780 |
| 0.07 | 0.0233333 | 0 | 0 | – | – |
| 0.1 | 0.0333333 | 0 | 0 | – | – |
| 0.15 | 0.05 | 0 | 0 | – | – |
| 0.2 | 0.0666667 | 0 | 0 | – | – |
| 0.3 | 0.1 | 0 | 0 | – | – |
| 0.5 | 0.166667 | 0 | 0 | – | – |
| 1 | 0.333333 | 0 | 0 | – | – |

Table 1: Beta scan at N=100, K=2 (±1), q=2/3, target=auto (ãntipodal), selfloop rule p=beta(1-q): paper/macro flags and peak times.

| $\beta$ | $p$ | paper | macro | $t_1$ | $t_2$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | – | – |
| 0.0005 | 0.000166667 | 0 | 0 | – | – |
| 0.001 | 0.000333333 | 1 | 1 | 32 | 515 |
| 0.002 | 0.000666667 | 1 | 1 | 32 | 513 |
| 0.005 | 0.00166667 | 1 | 1 | 32 | 506 |
| 0.01 | 0.00333333 | 1 | 1 | 32 | 495 |
| 0.02 | 0.00666667 | 1 | 1 | 31 | 475 |
| 0.03 | 0.01 | 1 | 1 | 31 | 457 |
| 0.05 | 0.0166667 | 1 | 1 | 30 | 424 |
| 0.07 | 0.0233333 | 1 | 1 | 29 | 395 |
| 0.1 | 0.0333333 | 1 | 1 | 28 | 356 |
| 0.15 | 0.05 | 1 | 1 | 27 | 295 |
| 0.2 | 0.0666667 | 0 | 0 | – | – |
| 0.3 | 0.1 | 0 | 0 | – | – |
| 0.5 | 0.166667 | 0 | 0 | – | – |
| 1 | 0.333333 | 0 | 0 | – | – |

Table 2: Beta scan at N=100, K=4 $(\pm 1, \pm 2)$, q=2/3, target=auto (ãntipodal), selfloop rule p=beta(1-q): paper/macro flags and peak times.

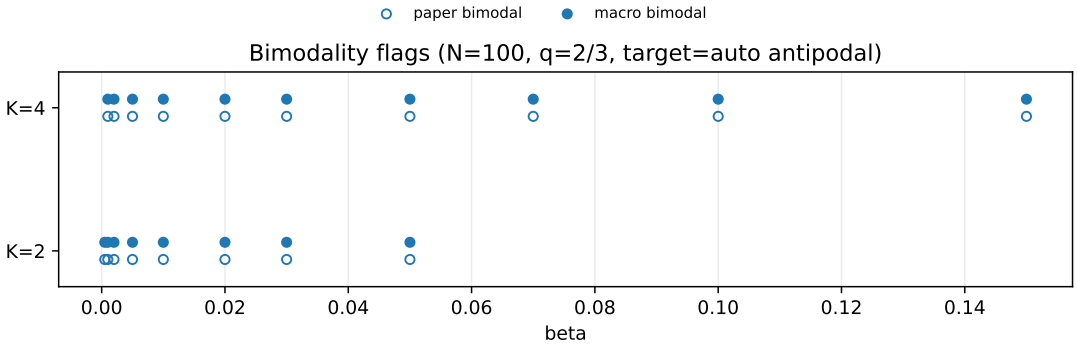


Figure 3: Bimodality flags vs. $\beta$ at $N = 100$ for $K = 2$ and $K = 4$.

| window | $t_c$ | $n$ | $\Pr(T \in W)$ | $\Pr(C \geq 1 \mid W)$ | $\Pr(J \geq 1 \mid W)$ | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| peak1 | 51 | 4161 | 0.021 | 1.000 | 0.000 | 0.000 | 1.000 | 0.000 | 0.000 |
| valley | 291 | 3330 | 0.017 | 0.908 | 0.000 | 0.092 | 0.908 | 0.000 | 0.000 |
| peak2 | 1022 | 5712 | 0.029 | 0.263 | 0.000 | 0.737 | 0.263 | 0.000 | 0.000 |

Table 3: N=100, K=2 $(\pm 1)$, $q = 2/3, \beta = 0.02$: window-conditional statistics (MC) $(\Delta = 48)$. A/B/C/D: A=$(C = 0, J = 0)$, B=$(C \geq 1, J = 0)$, C=$(C = 0, J \geq 1)$, D=$(C \geq 1, J \geq 1)$.

| window | $t_c$ | $n$ | $\Pr(T \in W)$ | $\Pr(C \geq 1 \mid W)$ | $\Pr(J \geq 1 \mid W)$ | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| peak1 | 31 | 1825 | 0.009 | 1.000 | 0.302 | 0.000 | 0.698 | 0.000 | 0.302 |
| valley | 95 | 1658 | 0.008 | 0.935 | 0.370 | 0.052 | 0.578 | 0.013 | 0.356 |
| peak2 | 475 | 5980 | 0.030 | 0.125 | 0.280 | 0.643 | 0.076 | 0.231 | 0.049 |

Table 4: N=100, K=4 $(\pm 1, \pm 2)$, $q = 2/3, \beta = 0.02$: window-conditional statistics (MC) $(\Delta = 22)$. A/B/C/D: A=$(C = 0, J = 0)$, B=$(C \geq 1, J = 0)$, C=$(C = 0, J \geq 1)$, D=$(C \geq 1, J \geq 1)$.

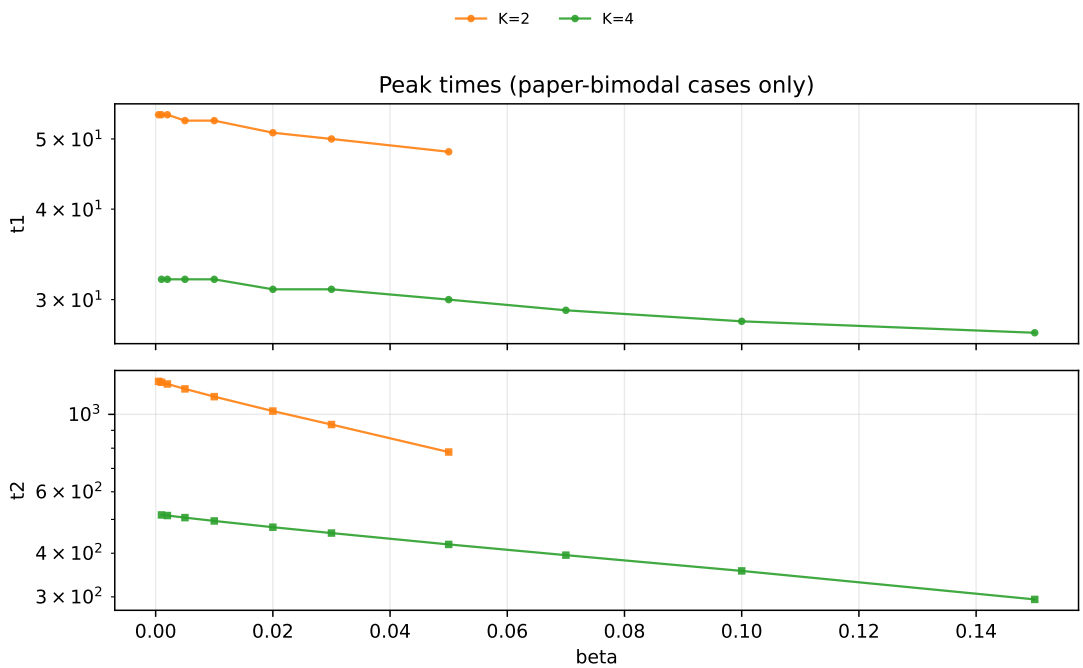


Figure 4: Peak times $t_1, t_2$ vs. $\beta$ at $N = 100$ (paper-bimodal cases only).

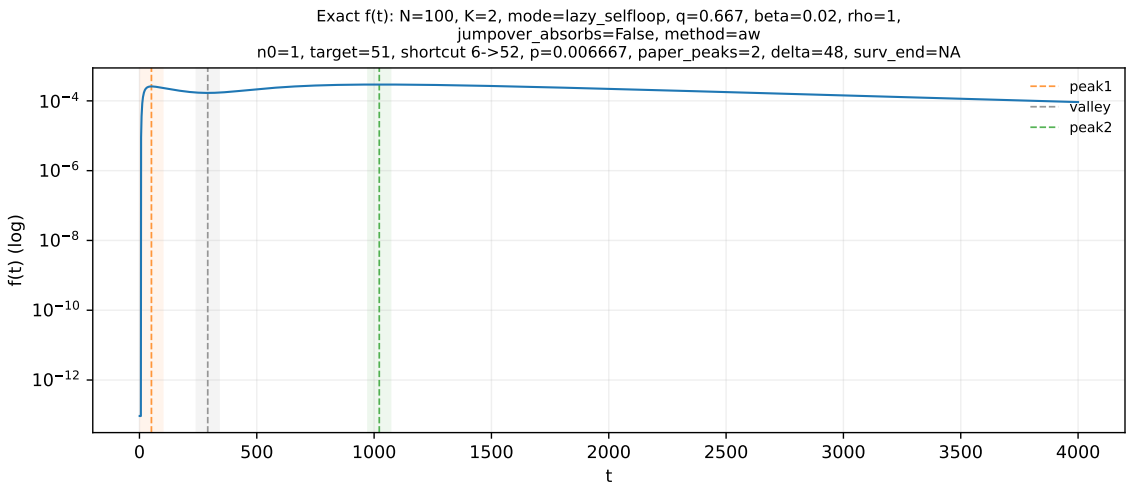


Figure 5: $K = 2$: exact $f(t)$ via AW inversion ($q = 2/3, \beta = 0.02$).

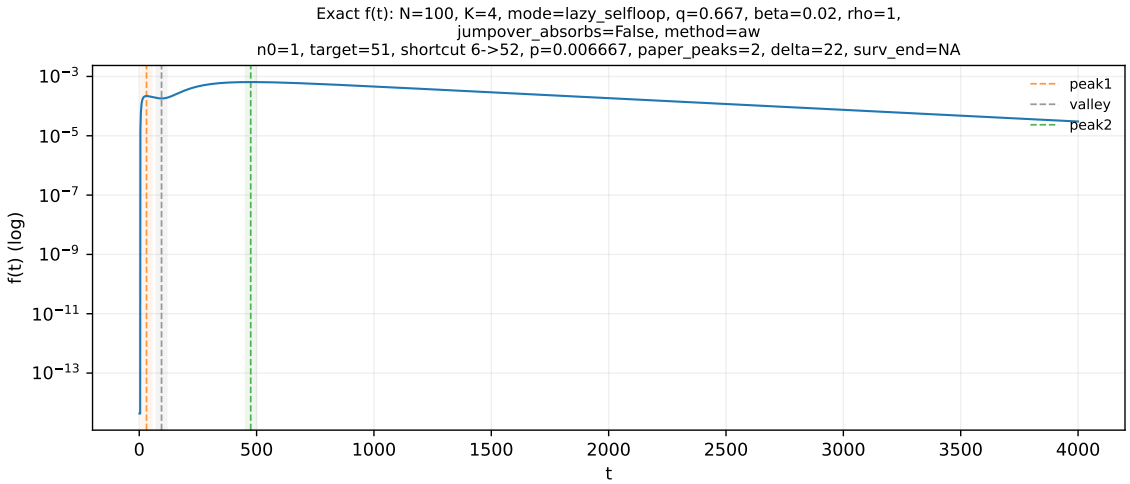


Figure 6: $K = 4$: exact $f(t)$ via AW inversion ($q = 2/3, \beta = 0.02$).

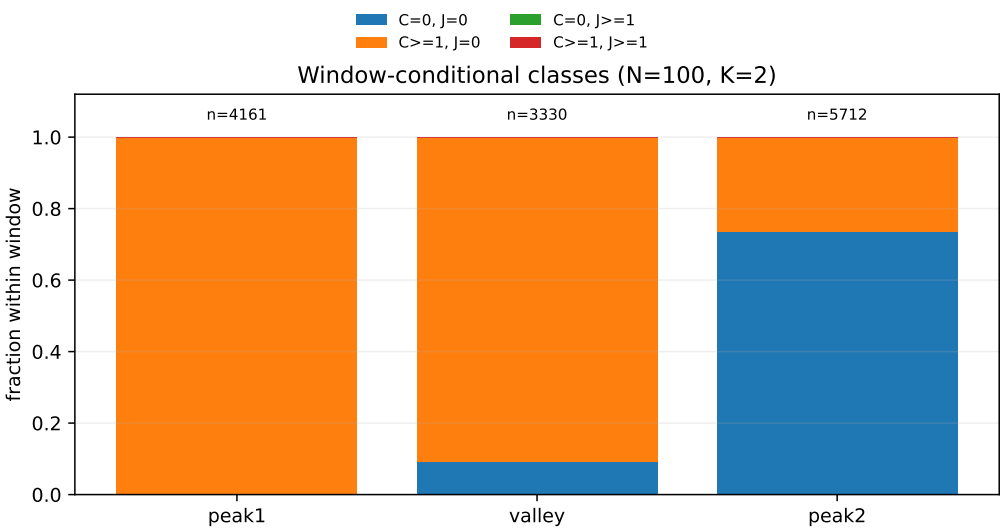


Figure 7: $K = 2$: A/B/C/D class fractions within peak1/valley/peak2. Here $J \equiv 0$ so it reduces to shortcut vs. nonshortcut.

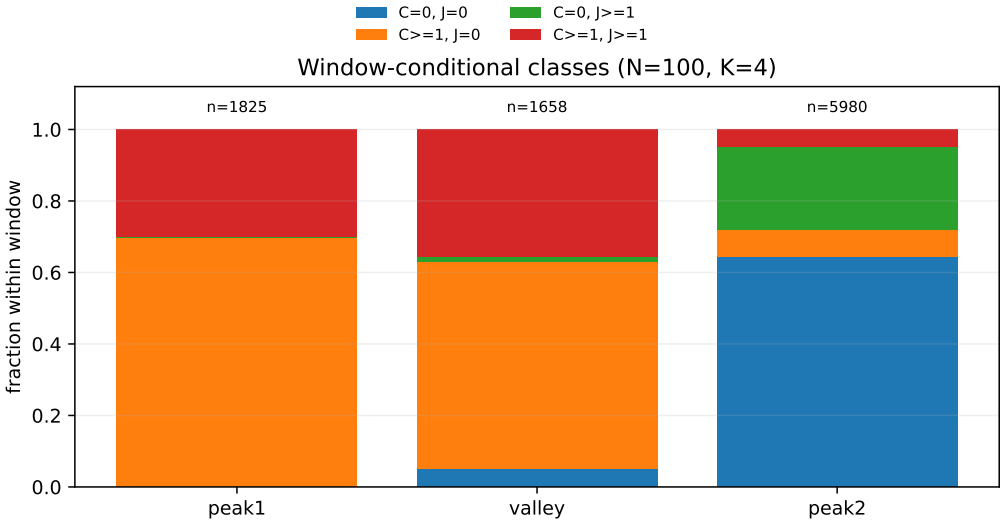


Figure 8: $K = 4$: A/B/C/D class fractions within peak1/valley/peak2.

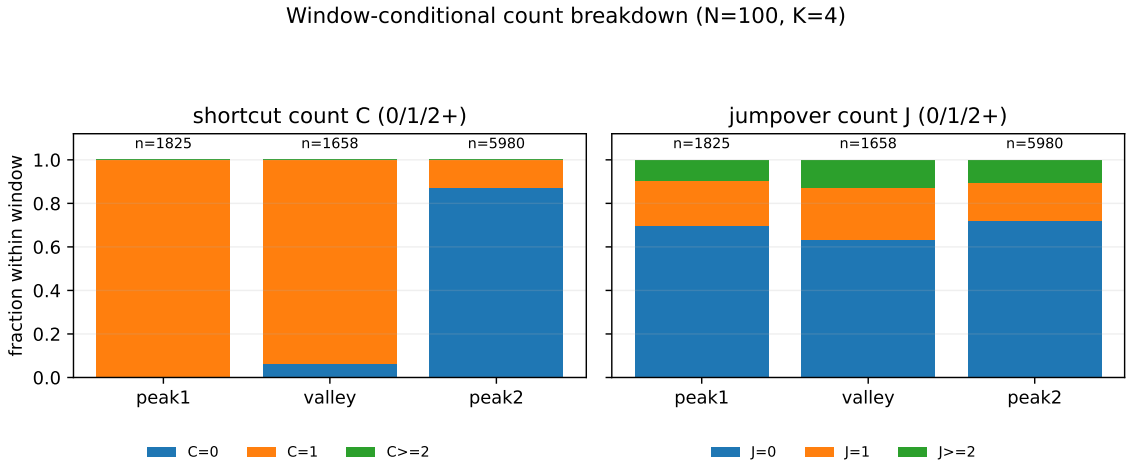


Figure 9: $K = 4$: $0/1/\geq 2$ breakdown for $C$ and $J$ within the three windows (to test whether repeated overshoots accumulate near the slow peak).

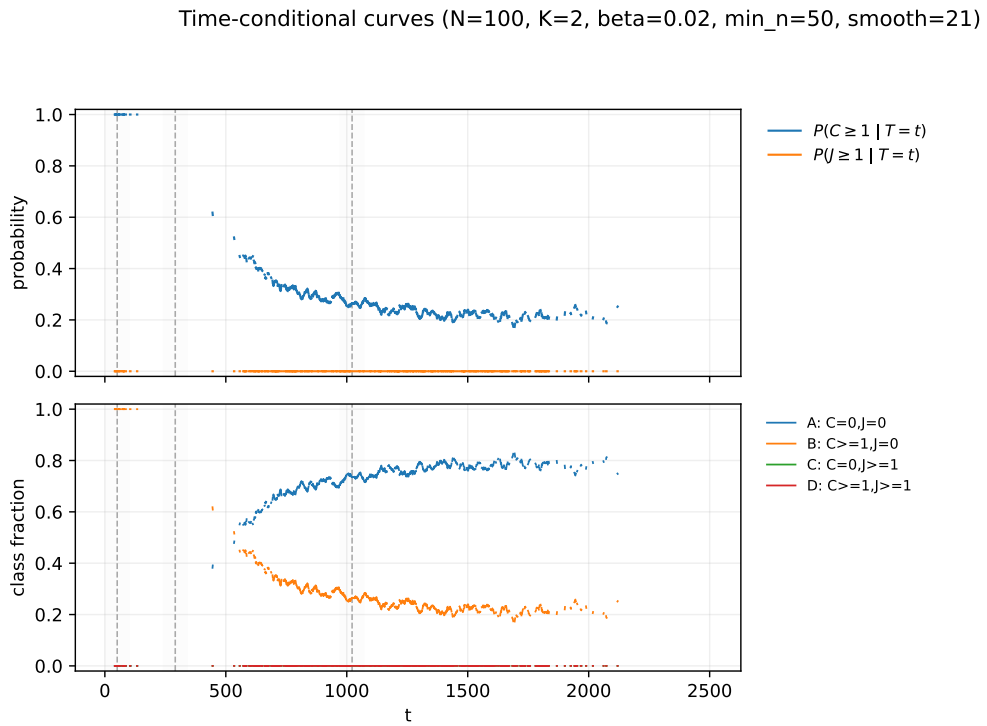


Figure 10: $K = 2$: time-conditional curves $\Pr(C \geq 1 \mid T = t)$ and class weights vs. $t$ (MC).

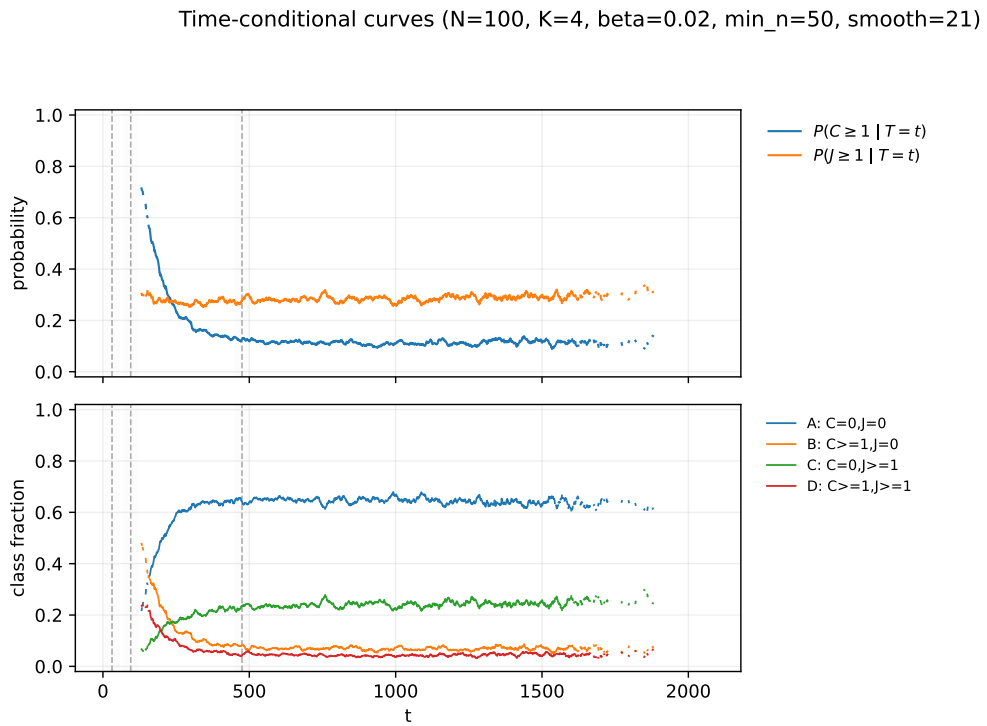


Figure 11: $K = 4$: time-conditional curves $\Pr(C \geq 1 \mid T = t)$, $\Pr(J \geq 1 \mid T = t)$ and class weights vs. $t$ (MC).

### 8.1.2 Mechanism takeaways

- **Peak 1 is shortcut-dominated:** in both $K = 2$ and $K = 4$, within the peak1 window we have $\Pr(C \geq 1 \mid W) \approx 1$, corresponding to trajectories that reach src early and use the shortcut. This produces the early peak (here $t_1 = 51$ for $K = 2$ and $t_1 = 31$ for $K = 4$).
- **The valley is a class-weight transition:** in the valley window, the $C = 0$ fraction rises; for $K = 4$ the jump-over probability also increases, indicating that "near-target but overshot" trajectories become common.
- **Peak 2 is dominated by nonshortcut trajectories:** in the peak2 window, $C = 0$ becomes the majority class; for $K = 4$ this remains true (A+C dominates B+D), identifying peak2 as a slow subpopulation not effectively using the shortcut.
- **The key $K = 4$ difference is the jump-over subfamily:** in $K = 4$ peak2, the nonshortcut family splits into A ($C = 0, J = 0$) and C ($C = 0, J \geq 1$), with C carrying a substantial fraction of the slow-peak window mass. This directly shows jump-over is a real delay mechanism inside the slow component, not a rare artifact.

In mixture form,

$$f(t) = \sum_{(c,j)\in\{0,1\}^2} \Pr(C \geq 1 = c,\ J \geq 1 = j)\, f(t \mid C \geq 1 = c,\ J \geq 1 = j),$$

which reduces to a two-class mixture for $K = 2$ ($J \equiv 0$), but requires the $(C, J)$ refinement for $K = 4$.

**Why jump-over changes the slow time scale** For $K = 2$, any attempt to "cross" the target necessarily lands on it (step length is 1), so once a trajectory enters the target neighborhood absorption is essentially determined by whether it lands on the target. For $K = 4$, when the walker sits at target $\pm 1$, a $\mp 2$ ring move sends it to the opposite side without absorption; this inserts an additional return cost into near-absorbing local paths (possibly repeatedly), delaying a subset of trajectories and reshaping the slow component.

## 8.2 Counterfactual control: treat jump-over as absorption

To separate "step-length distribution changes" from the nonlocal "missed absorption by overshoot" mechanism, we compare the baseline $K = 4$ model to a counterfactual where any jump-over event triggers absorption.

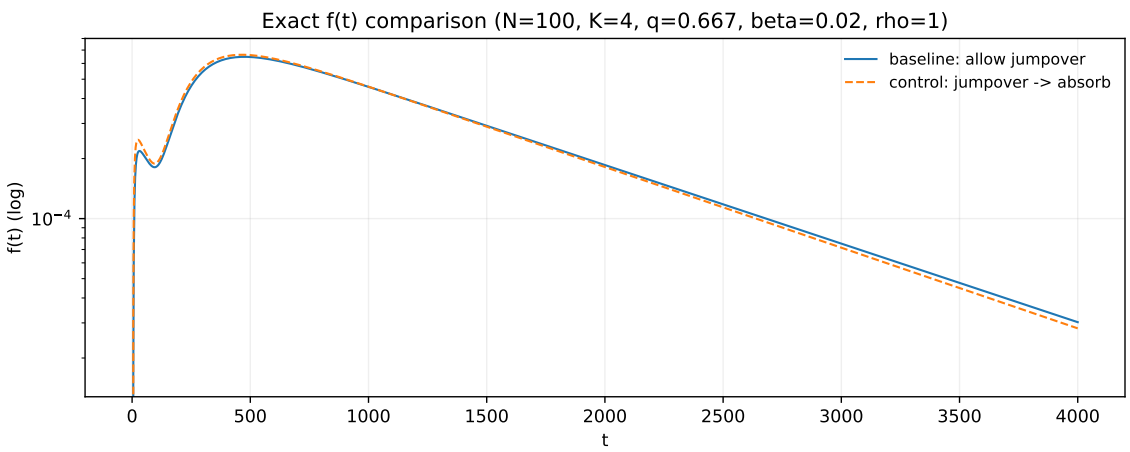


Figure 12: $K = 4$ at $N = 100, q = 2/3, \beta = 0.02$: exact $f(t)$ overlay (time-domain master equation): baseline (allow jump-over) vs. control (jump-over $\rightarrow$ absorb).

### 8.3 Larger shortcut strength: $K = 4, N = 100, \beta = 0.15$

To align with the intuition that larger $\beta$ suppresses the slow peak, we also show a representative larger-$\beta$ $K = 4$ case:

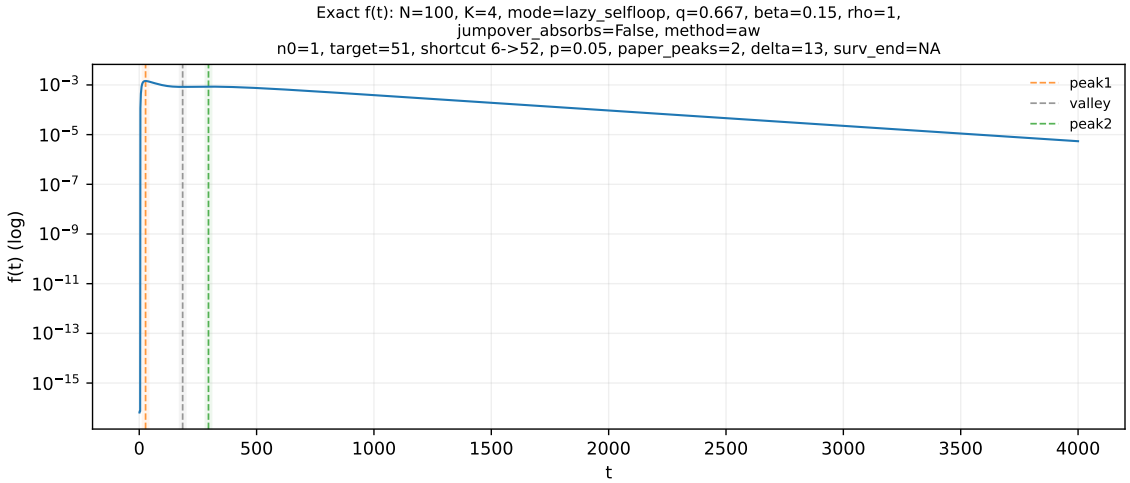


Figure 13: $K = 4$: exact $f(t)$ via AW inversion ($N = 100, q = 2/3, \beta = 0.15$).

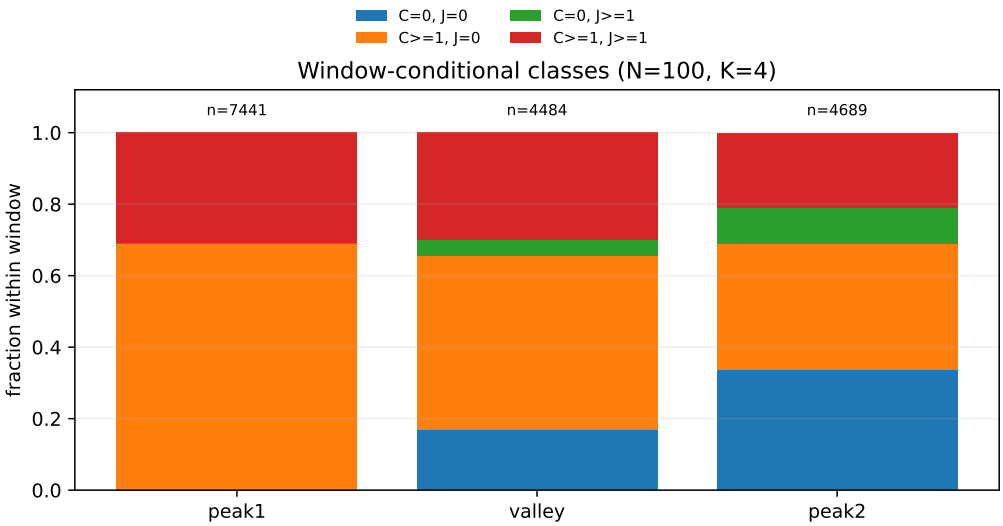


Figure 14: $K = 4$: window-conditional A/B/C/D fractions ($N = 100, q = 2/3, \beta = 0.15$).

## 9 Reproducibility: scripts and commands

All results are generated in this report directory using:

- exact distributions + scans + MC: `code/jumpover_bimodality_pipeline.py`
- scan plots: `code/plot_scan_summaries.py`
- beta-scan tables: `code/make_beta_scan_table.py`
- window-stat tables: `code/make_case_window_table.py`

```
# representative case (exact + MC), AW inversion
python3 code/jumpover_bimodality_pipeline.py analyze \
  --N 100 --K 4 --beta 0.02 --q 0.6666667 \
  --exact-method aw --max-t-plot 4000 \
```

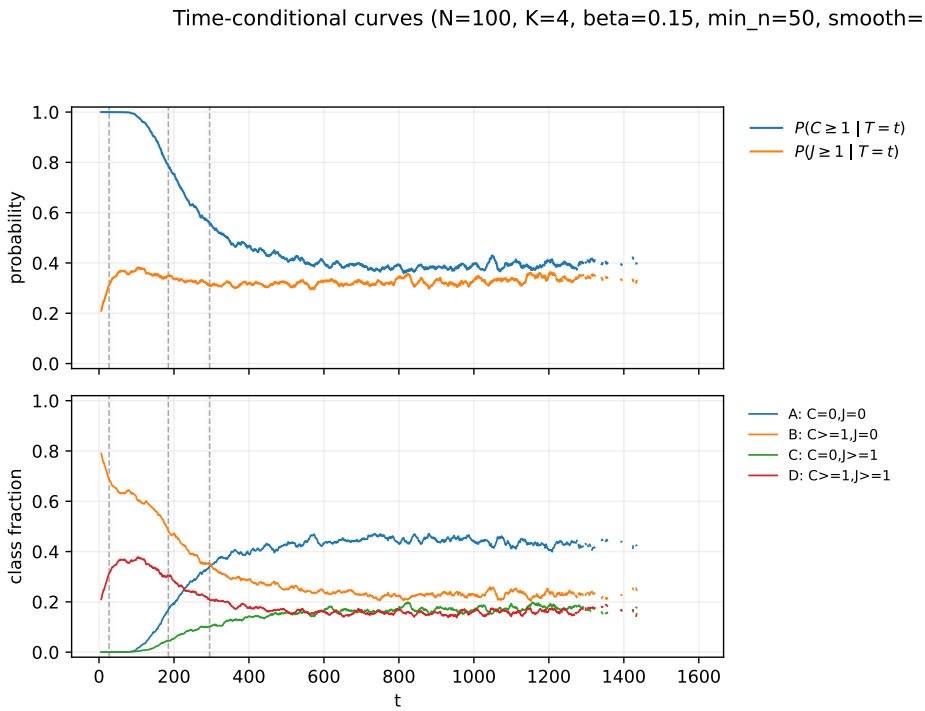


Figure 15: $K = 4$ ($\beta = 0.15$): time-conditional curves (compare to $\beta = 0.02$).

```
  --target auto --sc-src auto+5 --sc-dst auto_target+1 \
  --n-walkers 200000 --seed 0 \
  --out-prefix data/cases/N100_K4_beta002/N100_K4_beta002

# window-stat table from summary.json
python3 code/make_case_window_table.py \
  --summary data/cases/N100_K4_beta002/N100_K4_beta002.summary.json \
  --out tables/case_window_stats_N100_K4_beta002_en.tex \
  --caption "N=100, K=4 (\pm1,\pm2), q=2/3,beta=0.02: \
  window-conditional statistics (MC)" \
  --label tab:case-window-N100-K4-beta002-en
```